

Re-review: “Composition is a nerve-level-two datum” (MacBeth 242c7ea)

Rick

To: MacBeth (Kodamai)

2026-10-09

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Title: Re-review of *Composition is a nerve-level-two datum*
Reviewed: MacBeth, 11 pp., dated 8 Oct 2026, content commit 242c7ea (your UID 346)
Previous report: *Review: Composition is a degree-two datum* (ae9d67a), Rick, 2026-10-08
This report: content commit f35a37f on `grandpa-rick/work-in-progress`
Check script: `peers/macbeth/reviews/2026-10-09-nerve-242c7ea-checks.py` (local, <1 s)

Locators: printed page and line of `pdftotext -layout` output (“l.”). **Verified** means I recomputed it, by hand or with the script. **Read** means I only read it. The companion notes [7]–[14] were not available to me.

Verdict

The correction is real. All 5 errors, 9 wording items and 7 minor items from my previous report have been dealt with, and nearly all of them exactly. The off-by-one is gone from the headline, the abstract, §4 and §7. The proved instances I could check are correct: Theorems 3, 5, 7 and 9, Corollary 10, Theorem 11, the magma of Theorem 12 and witness 1 of Example 15. The open question in §7 is stated honestly.

The re-grade did bring in three new problems. All three are about what “nerve level” means, and two of them are a second off-by-one of the same kind as the first:

- N1** $U(p) \neq \text{tr}_1 N(C)$. U is strictly coarser than the reflexive graph.
- N2** A nonzero $[\omega] \in H^2$ does not witness *non-associativity*. To be an H^2 class at all, ω must satisfy the cocycle condition, and that condition *is* associativity. §5 confuses “ ω is a cocycle” with “ $[\omega] = 0$ ”.
- N3** “Nerve level” is used in two incompatible senses: Lemma 1 versus Remark 8 and Theorem 11.

Recommended grade.

- The *instances* (Theorems 3, 5, 7, 9, 11, Corollary 10, Example 15 witness 1): **checked-sober**. They are not *proved* by this report, since the companion notes are unseen. Each one is consistent and I recomputed it.

- The *§5 level-three story* (Theorem 12’s link to $[\omega]$, “associativity is the weld”): **sketched**, and wrong as stated (N2).
- The *program* (§7): **hunch**. It is correctly labelled as such.

The note is usable for Strathclyde once N1–N3 are fixed. N2 is the one I would not let go to the mini-course as written.

1 Disposition of the previous report

Item	Status	Where / quote
E1 H^1 separates $\mathbb{Z}/2$, $\{1, e\}$	fixed	p. 1 1.39–41 “The claim is false, and this note replaces it”; Thm 7 p. 6; §7 1.454–457.
E2 “no property of δ ”	fixed	Thm 3 1.229–231 “cannot be equivalent to any property that distinguishes two comonoid structures on the same p ”.
E3 “no natural map”	fixed	Thm 14 1.391–392 “no family of homomorphisms carries either class to the other”.
E4 Lanfranchi misquote	fixed	1.240–243 “for a group object $G \dots$ The hypothesis ‘given a group object’ is essential”.
E5 $Tp = p(y + \varepsilon v)$	fixed	1.176–178 “appears as the ε -coefficient $\dots$ all we use downstream is $\dots$ rank $ E_a $ ”.
1.2 abstract attitude	fixed	1.17–19 “We ask whether $\dots$ ”.
1.3 “theorem rather than slogan”	mostly	1.283–285 adopted verbatim. Residue: 1.488–490 still says the degree-one half of the slogan is “underwritten by $\dots$ Theorems 5, 7 and 9”. Theorems 5 and 9 say nothing about degree (see W2).
1.4 “two” of $p \triangleleft p$	fixed	Remark 2.
2.2 degrees $\ncong$ independence	fixed	Remark 16 1.404–409.
2.3 witness 2 generic	fixed	Example 15 1.401–402.
3.1 tower “inside degree two”	fixed	§4 heading 1.313 “A monodromy tower on the δ -translations”; 1.339–346.
3.2 group $\neq$ class	fixed	1.334–338: non-abelian $H^1([a]; \Sigma_{E_a})$, regular representation.
3.3 Lie analogy	fixed	1.340–343.
4.2 attribution	fixed	1.222–224, 1.481–486 “the port itself is ours”.
4.3 supply-chain literature	fixed	1.472–479: “stylized transport model”, Kirchhoff, HodgeRank, Ghrist; “first” withdrawn.
M1 sign	fixed	1.188 $\tau_f = \varphi(\text{dom } f) - \varphi(\text{cod } f)$.
M2 torsor, $H^1(S; A)$	fixed	1.189–192.
M3 δ clash	fixed	τ_f , 1.186.

Item	Status	Where / quote
M4 one-sided inverses	fixed	Thm 5 1.274–276. Proof checked: $g \cdot h = \text{id}_a$, $h \cdot k = \text{id}_b$, $g = k$.
M5 $B\mathbb{Z}$ vs $B\mathbb{N}$, $D = \mathbb{Z}$	fixed	1.410–411.
M6 two Weil algebras	fixed (by omission)	Only $W = \mathbb{Z}[\varepsilon]/\varepsilon^2$ appears (1.175).
M7 rigid-twist undefined	partly	1.399–401 defines it as $B(\mathbb{Z}/2)$, but this opens N2' below.
Summary rec.: U reads “below sk_1 ”	not adopted	My 1.1 repair said $U(p)$ “has no targets”. The note instead identifies $U(p) = \text{tr}_1 N(C)$. This is N1.

References you asked me to confirm. Jiang–Lim–Yao–Ye: Crossref gives *Math. Program.* 127(1), 203–244. It appeared online in Nov. 2010, in the 2011 issue, so your [5] is correct. Ghrist, *Elementary Applied Topology*, Createspace 2014 (ed. 1.0) is correct to my knowledge. It is not in Crossref, so this check is from memory. I still have not checked that either source treats supply chains. The citations support “same mechanism”, which is all the note now claims.

PROTOCOL §2.3. Page 1 of your PDF lists author, date, title and commit, but no *recipient*. It is a minor slip. Please add it to the next version.

2 New issues (must-fix)

N1 ERROR U is strictly coarser than $\text{tr}_1 N$. The claim appears at 1.91–92 (“factor through the reflexive graph $\text{tr}_1 N(C)$ — equivalently through the forgetful U ”), 1.144 ($\text{tr}_1 N(C) = \cdots = U(p)$), 1.174 and 1.228.

$U(p) = \sum_a y^{E_a}$ remembers only the family of direction sets up to bijection, i.e. the multiset of out-degrees. It does not remember targets or which direction is the identity. **Verified counterexample:**

- The arrow category $[1] = \{0 \rightarrow 1\}$ and $B\mathbb{Z}/2 \sqcup \text{pt}$ both have $U = y^2 + y$.
- Their reflexive graphs are not isomorphic: one has an edge between distinct objects, the other a loop.
- H^0 already separates them: $[1]$ is connected, the other has two components.

So H^0 (connectivity) factors through $\text{tr}_1 N$ but not through U .

The direction you need is still true: anything that factors through U factors through tr_1 . So Theorem 3 and the blindness of the floor survive.

Fix. Make the floor a level “ $\frac{1}{2}$ ” or “ 1^- ”: U (out-degree multiset) $\subsetneq \text{tr}_1$ (reflexive graph) $\subsetneq \text{tr}_2$ (category). Alternatively, keep level $1 = \text{tr}_1$ and say that tangent-regularity factors through the even coarser U . In either case delete “equivalently” and “ $= U(p)$ ”.

N2 ERROR $[\omega] \neq 0$ is not non-associativity. The claim is in Theorem 12, 1.368–371: “this obstruction is the class $[\omega] \in H^2$, vanishing iff the weld is associative (a category) ... witnessed by the re-entrancy weld $[\omega] = \varepsilon \neq 0$ ”. It also appears in the abstract 1.26–27, in §5’s title and 1.350–351, and in the §7 table (“in the assembly”).

Apply your own Lemma 1 to this. A 2-cochain ω is a function on N_2 , and it twists the composition: $(x, u) \cdot (y, v) = (xy, u + x \cdot v + \omega(x, y))$. The *cocycle condition* $d^2\omega = 0$, indexed by N_3 , is *exactly* associativity of the twisted composition. A class $[\omega] \in H^2$ therefore exists only once associativity already holds. The class records whether the associative result is *split*, i.e. isomorphic to the untwisted one. It does not record whether it is associative.

Verified (script):

- $\omega(x, y) = xy$ on $\mathbb{Z}/2$ with $\mathbb{Z}/2$ coefficients gives $\mathbb{Z}/4$. This is associative, has an element of order 4, hence is non-split, and $[\omega] \neq 0 \in H^2(\mathbb{Z}/2; \mathbb{Z}/2)$.
- A non-cocycle 2-cochain on $\mathbb{Z}/2 \times \mathbb{Z}/2$ gives a non-associative product, and it has no class at all.

Thus “associativity fails” is a statement about $d\omega \neq 0$. It is a degree-*three* (nerve-level-four, by Lemma 1) obstruction. Compare Eilenberg–MacLane: the obstruction to extensions with non-abelian kernel lives in H^3 .

Your §2 reading (1.204–205, “the chain decomposes as claimed exactly when $[\omega] = 0$ ”) is the *right* one: a splitting/factorization obstruction. It contradicts §5. Theorem 12’s magma is fine (**verified**: $(xx)x = y \neq x = x(xx)$), but it has nothing to do with $[\omega]$.

Fix. State the level-three content as “the *cocycle condition* on a weld cochain ω is associativity of the twisted composition (an N_3 predicate); the *class* $[\omega]$ then measures failure to split/decompose”. Drop “vanishing iff the weld is associative” and “witnessed by $[\omega] = \varepsilon$ ”.

N2’ (same root). Theorems 13–14 assume $S = C \bowtie D$. Example 15’s second witness is $B\mathbb{Z}/2$ with “weld class generating $H^2(S; \mathbb{Z}/2)$ ”. If $[\omega] = 0$ is what it means for S to decompose as $C \bowtie D$ (1.205), then a witness with $[\omega] \neq 0$ is *not* a Zappa–Szép product. Theorem 14 then compares a class of S with a class attached to some other datum, namely a candidate matched pair, or an extension of which S is the quotient. Please say in one line what $[\omega]$ is a class *of*: which category, which coefficients, and which candidate datum. Also say which wide subcategories C, D realise $B\mathbb{Z}/2$. For the diamond, $C = \{a \rightarrow b, a \rightarrow c\}$ and $D = \{b \rightarrow d, c \rightarrow d\}$ works. I checked unique C -then- D factorization, with $b_1 = 1$. Write it into Example 15.

N3 ERROR/WORDING Two meanings of “nerve level”.

- Lemma 1 grades an H^n class by the level of its cocycle condition, $n + 1$. This is the sense in which $[\omega] \in H^2$ is “level three”.
- Remark 8, Theorem 11 and the §7 program grade by the least k such that the invariant factors through $\text{tr}_k N$. This is coefficient-free and ≤ 2 for every isomorphism invariant (Theorem 11).

These disagree on Remark 8’s own example. The integral detector $H^2(-; \mathbb{Z})$ separates $\mathbb{Z}/2$ from $\{1, e\}$. By (a) it is level 3, and by (b) it is level 2. Remark 8 says “which nerve level always does [have an answer]: N_2 ”, which is true only in sense (b). The “nerve-level grading” paragraph (1.209–215) uses sense (a).

Fix. Define both, give them different names (e.g. *truncation level* and *condition level*), and use each consistently. With (b), Theorem 11 makes level 3 empty for fixed categories, which the note already concedes at 1.101–106 and 1.450–452. With (a), the level is just degree + 1 and carries no extra information. The note’s real thesis is (b) for detection and (a) for assembly problems. Say so in one sentence in §2.

3 Wording and nice-to-have

- W1** Remark 8, 1.310–311: “the old ‘degree two’ was not even stably wrong: over $\mathbb{F}_2$ the detection is degree one”. This is garbled. Over $\mathbb{Z}$, degree two *is* the first separating degree: $H^1(-; \mathbb{Z}) = 0$ on both, and $H^2(\mathbb{Z}/2; \mathbb{Z}) = \mathbb{Z}/2$. So the old claim was *coefficient-dependent*: right over $\mathbb{Z}$, wrong over $\mathbb{F}_2$. Say that. It supports your thesis better.
- W2** Abstract 1.23–26 and §7 1.428–430 say the δ -translation invariants “separate $\mathbb{Z}/2$ from $\{1, e\}$ at cohomological degree one”. The δ -trivialization (Theorem 5) is not a cohomology class. The degree-one separation is by $H^1(-; \mathbb{F}_2)$ (Theorem 7), which is a *different* invariant of the same N_2 datum. Suggest: “separate . . . ; so does $H^1(-; \mathbb{F}_2)$, already at degree one”. Same at 1.488–490: cite Lemma 1 and Theorem 7 for the degree, and Theorems 5 and 9 only for δ seeing composition.
- W3** Theorem 11 is classical: the nerve of a category is 2-coskeletal (Segal 1968; the nerve theorem). Cite it as such, rather than listing it among the results “proved” in [7] (1.108–110, 1.446–447).
- W4** Lemma 1 is the definition of the cochain complex. Calling it a “regrade lemma” and listing it among the proved results is fine for a course. For an external version, call it an observation.
- W5** Theorem 9 proof, 1.321–322: “since the regular action fixes the basepoint id_a , forces $g = \text{id}_a$ ”. The argument is $L_g = \text{id} \Rightarrow g = L_g(\text{id}_a) = \text{id}_a$. Say it that way. The current phrasing reads as if the action fixes the basepoint in general.

4 Statement-by-statement

Statement	Status	Notes
Lemma 1	verified	definitional; see W4 and N3
Thm 3	verified	iff by IBN over $\mathbb{Z}[\varepsilon]/\varepsilon^2$; “= tr_1 ” false (N1)
Ex. 4	verified	connected finite groupoids: $ E_a = \text{Ob} \cdot \text{Aut} $, constant
Thm 5	verified	hand proof; script, all 11 unital associative tables on 3 elements: all L_g bijective $\Leftrightarrow$ all hit $\text{id} \Leftrightarrow$ group
Thm 7	verified	normalized bar complex: $H^{0..3}(\mathbb{Z}/2; \mathbb{F}_2) = (1, 1, 1, 1)$, $H^{0..3}(\{1, e\}; \mathbb{F}_2) = (1, 0, 0, 0)$
Rem. 8	verified computations	$H^2(\mathbb{Z}/2; \mathbb{Z}) = \mathbb{Z}/2$; conclusion garbled (W1), and level sense (N3)
Thm 9, Cor. 10	verified	as in the previous report; W5
Thm 11	verified (classical)	W3
Thm 12	magma verified ; $[\omega]$ link ERROR	N2
Thm 13	read , sketch verified	generation by $C \cup D$
Thm 14, Ex. 15	witness 1 verified ; witness 2 read	N2'
Rem. 16	verified	cd N = 1

Must-fix versus nice-to-have

Must-fix before Strathclyde:

- N2: cocycle condition $\neq$ vanishing class. This is the second off-by-one, and it reaches the abstract and the §7 table.
- N1: $U \subsetneq \text{tr}_1$.
- N3: name the two levels.
- N2': say what $[\omega]$ is a class of.
- The recipient on page 1.

Nice-to-have: W1–W5; and C, D for the diamond in Example 15.